Antiderivatives and Riemann Sums

Basic antiderivatives

1 Find the general antiderivative of each function:

a $f(x) = x^4$

b $f(x) = \frac{1}{x}$

c $f(x) = e^x$

d $f(x) = \sec^2 x$

2 Find $\int (3x^2 - 4x + 5)\,dx$.

3 Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.

Riemann sums: left, right, and midpoint

4 Approximate $\int_0^4 x^2\,dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.

5 Approximate $\int_0^4 x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

6 Approximate $\int_0^4 x^2\,dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.

The Trapezoidal Rule

7 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2\,dx$.

8 A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x)\,dx$.

Sums as limits: the definite integral

9 Explain how the definite integral $\int_a^b f(x)\,dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.

10 Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2\,dx$, and explain why using the shape of the graph.

Riemann sums from a table of values

11 A table gives $f(0)=3$, $f(1)=5$, $f(2)=8$, $f(3)=10$, $f(4)=9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

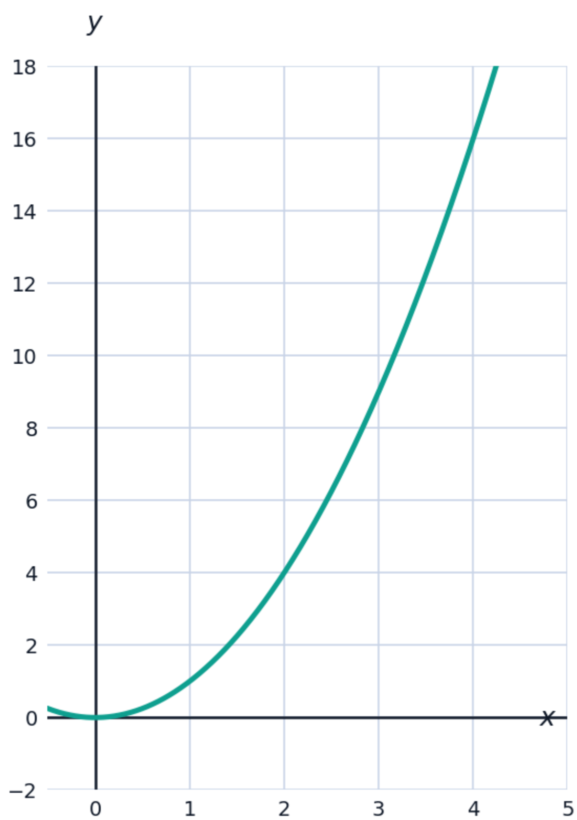
Antiderivatives of trigonometric and exponential functions

12 Find $\int (3\sin x + 2e^x)\,dx$.

13 Find the particular antiderivative $F(x)$ of $f(x) = \sec^2 x$ satisfying $F(0) = 2$.

Visualizing a Riemann sum approximation

14 The graph shows $f(x) = x^2$ on $[0, 4]$. Use the graph to explain why the Right Riemann Sum $R_4 = 30$ (found earlier) overestimates the true area, and why the Left Riemann Sum $L_4 = 14$ underestimates it.



An applied antiderivative problem

15 A factory's production rate is $r(t) = 100 + 20t$ units/day, where t is days since a process change. Find the total production function $P(t)$ given $P(0) = 0$, then find the total production over the first 10 days.

More antiderivative practice

16 Find the general antiderivative of $f(x) = 4x^3 - \frac{2}{x^2} + \cos x$.

17 Find the particular antiderivative $F(x)$ of $f(x) = x^2 - 3$ satisfying $F(0) = 4$.

Comparing Riemann sums to the exact integral

18 For $\int_0^4 x^2\,dx$, the exact value is $\frac{64}{3} \approx 21.33$. Rank the four approximations found earlier — $L_4 = 14$, $R_4 = 30$, $M_4 = 21$, $T_4 = 22$ — from closest to farthest from the exact value, and explain which type of approximation (midpoint or trapezoidal) tends to be more accurate for a smooth curving function.